

SHIVANSH PRATAP SINGH

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PUBLICATIONS

AIoT Health: Medicine Reminder For The Elderly (Primary Author)

2025

Accepted at ICDECT 2025

(to appear in Springer LNNS)

- **Designed a two-stage cascading pipeline** on an ESP32-S3 using a Keyword Spotting (KWS) and an Spoken Language Understanding (SLU) classifier (Intent Recognition). With an optimized memory footprint by quantisation and manual tensor arena allocation, fitting the pipeline into 330KB of usable SRAM. Achieved 96.4% KWS accuracy and 94.1% SLU with 100ms end-to-end latency on a 15 hardware bill-of-materials.
- **Preprint:** 10.5281/zenodo.19034554

EXPERIENCE

Cadence Design Systems

Feb 2024 – Feb 2025

Software Engineer I

Noida

- Maintained strict system invariants across file I/O, data models, and Qt-based UI layers for libraries, ensuring fast load times.
- Engineered a localized data-consistency layer for the DE-HDL flow, enabling chip designers to perform symbol operations (rename, duplicate, delete) without server-side Pulse connectivity; each operation required full consistency across the symbol, footprint, part, and property layers in DE-HDL file format
- Optimized symbol load pipelines by implementing efficient metadata caching and specialized data structures to handle high-concurrency operations in local mode.
- Collaborated with architects and principal engineers to ship production-ready features into Allegro OrCAD X.
- Authored unit and regression tests in Tcl; worked in Linux environments with Perforce, internal CI/CD pipelines, and regression frameworks

Datacouch.io

Nov 2022 – March 2023

Intern

Remote

- Built workflow automation systems using Python and third-party APIs; developed and evaluated image classification ML pipelines; produced architecture documentation for client deliverables

EDUCATION

University of Petroleum and Energy Studies (UPES)

2019 – 2023

B.Tech, Computer Science and Engineering — Honours in DevOps

Dehradun, India

CGPA: 7.77/10

RESEARCH INTERESTS

Investigating hardware-software co-design methodologies to enable robust, on-device model adaptation within strict memory and power envelopes. Specifically focused on bridging the generalization gap in TinyML through hardware-constrained architecture search and efficient tensor management.

CERTIFICATIONS

Digital Systems: From Logic Gates to Processors: Universitat Autònoma de Barcelona [Coursera];

VLSI CAD Part I: Logic: University of Illinois Urbana-Champaign [Coursera]

Linear Circuits 2: AC Analysis: Georgia Institute of Technology [Coursera]

C++ For C Programmers Part A and B: University of California, Santa Cruz [Coursera];

SKILLS

Systems Programming: C++ (11/14/17), C, Python, Tcl, Qt Framework

TinyML & Embedded: TensorFlow Lite Micro, ESP-IDF (PSRAM/SRAM Management), RTOS, I²S/I²C/UART, Edge Impulse

Hardware & EDA: Verilog (Basic Synthesis), Logic Design Fundamentals, DE-HDL Flow, Allegro X Infrastructure

Developer Tools: Advanced Linux, Git, Perforce (Industrial VCS), Docker, CI/CD Pipelines